

Homogeneous Transformation Matrices from Base to Joint

T_01

$$\begin{pmatrix} \cos(\theta_1) & -\sin(\theta_1) & 0 & 90 \\ \sin(\theta_1) & \cos(\theta_1) & 0 & 0 \\ 0 & 0 & 1 & -\frac{591}{2} \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

T_10

$$\begin{pmatrix} \cos(\theta_1) & -\sin(\theta_1) & 0 & 90 \\ \sin(\theta_1) & \cos(\theta_1) & 0 & 0 \\ 0 & 0 & 1 & -\frac{591}{2} \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

T_02

$$\begin{pmatrix} -\cos(\theta_1) \sin(\theta_2) & -\cos(\theta_1) \cos(\theta_2) & \sin(\theta_1) & 200 \cos(\theta_1) + 90 \\ -\sin(\theta_1) \sin(\theta_2) & -\cos(\theta_2) \sin(\theta_1) & -\cos(\theta_1) & 200 \sin(\theta_1) \\ \cos(\theta_2) & -\sin(\theta_2) & 0 & -\frac{591}{2} \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

T_21

$$\begin{pmatrix} -\sin(\theta_2) & -\cos(\theta_2) & 0 & 200 \\ 0 & 0 & -1 & 0 \\ \cos(\theta_2) & -\sin(\theta_2) & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

T_03

$$\begin{pmatrix} \sin(\theta_1) \sin(\theta_3) - \cos(\theta_1) \cos(\theta_3) \sin(\theta_2) & \cos(\theta_3) \sin(\theta_1) + \cos(\theta_1) \sin(\theta_2) \sin(\theta_3) & \cos(\theta_1) \cos(\theta_2) & 200 \cos(\theta_1) + \frac{305 \cos(\theta_1) \cos(\theta_2)}{2} \\ -\cos(\theta_1) \sin(\theta_3) - \cos(\theta_3) \sin(\theta_1) \sin(\theta_2) & \sin(\theta_1) \sin(\theta_2) \sin(\theta_3) - \cos(\theta_1) \cos(\theta_3) & \cos(\theta_2) \sin(\theta_1) & 200 \sin(\theta_1) + \frac{305 \cos(\theta_2) \sin(\theta_1)}{2} \\ \cos(\theta_2) \cos(\theta_3) & -\cos(\theta_2) \sin(\theta_3) & \sin(\theta_2) & \frac{305 \sin(\theta_2)}{2} - \frac{591}{2} \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

T_32

$$\begin{pmatrix} \cos(\theta_3) & -\sin(\theta_3) & 0 & 0 \\ 0 & 0 & -1 & -\frac{305}{2} \\ \sin(\theta_3) & \cos(\theta_3) & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

T_04

$$\begin{pmatrix} \cos(\theta_4) (\sin(\theta_1) \sin(\theta_3) - \cos(\theta_1) \cos(\theta_3) \sin(\theta_2)) + \sin(\theta_4) (\cos(\theta_3) \sin(\theta_1) + \cos(\theta_1) \sin(\theta_2) \sin(\theta_3)) & \cos(\theta_4) (\cos(\theta_3) \sin(\theta_1) + \cos(\theta_1) \sin(\theta_2) \sin(\theta_3)) \\ -\cos(\theta_4) (\cos(\theta_1) \sin(\theta_3) + \cos(\theta_3) \sin(\theta_1) \sin(\theta_2)) - \sin(\theta_4) (\cos(\theta_1) \cos(\theta_3) - \sin(\theta_1) \sin(\theta_2) \sin(\theta_3)) & \sin(\theta_4) (\cos(\theta_1) \sin(\theta_3) + \cos(\theta_3) \sin(\theta_1) \sin(\theta_2)) \\ \cos(\theta_2) \cos(\theta_3) \cos(\theta_4) - \cos(\theta_2) \sin(\theta_3) \sin(\theta_4) & 0 \\ 0 & 0 \end{pmatrix}$$

T_43

$$\begin{pmatrix} \cos(\theta_4) & -\sin(\theta_4) & 0 & 0 \\ \sin(\theta_4) & \cos(\theta_4) & 0 & 0 \\ 0 & 0 & 1 & \frac{457}{2} \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

T_05

$$\begin{pmatrix} \cos(\theta_4) (\sin(\theta_1) \sin(\theta_3) - \cos(\theta_1) \cos(\theta_3) \sin(\theta_2)) + \sin(\theta_4) (\cos(\theta_3) \sin(\theta_1) + \cos(\theta_1) \sin(\theta_2) \sin(\theta_3)) & -\cos(\theta_1) \cos(\theta_2) & \cos(\theta_4) \\ -\cos(\theta_4) (\cos(\theta_1) \sin(\theta_3) + \cos(\theta_3) \sin(\theta_1) \sin(\theta_2)) - \sin(\theta_4) (\cos(\theta_1) \cos(\theta_3) - \sin(\theta_1) \sin(\theta_2) \sin(\theta_3)) & -\cos(\theta_2) \sin(\theta_1) & \sin(\theta_4) \\ \cos(\theta_2) \cos(\theta_3) \cos(\theta_4) - \cos(\theta_2) \sin(\theta_3) \sin(\theta_4) & -\sin(\theta_2) & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

T_54

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 118 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

T_06

$$\begin{pmatrix} \cos(\theta_4) (\sin(\theta_1) \sin(\theta_3) - \cos(\theta_1) \cos(\theta_3) \sin(\theta_2)) + \sin(\theta_4) (\cos(\theta_3) \sin(\theta_1) + \cos(\theta_1) \sin(\theta_2) \sin(\theta_3)) & -\cos(\theta_1) \cos(\theta_2) & \cos(\theta_4) \\ -\cos(\theta_4) (\cos(\theta_1) \sin(\theta_3) + \cos(\theta_3) \sin(\theta_1) \sin(\theta_2)) - \sin(\theta_4) (\cos(\theta_1) \cos(\theta_3) - \sin(\theta_1) \sin(\theta_2) \sin(\theta_3)) & -\cos(\theta_2) \sin(\theta_1) & \sin(\theta_4) \\ \cos(\theta_2) \cos(\theta_3) \cos(\theta_4) - \cos(\theta_2) \sin(\theta_3) \sin(\theta_4) & -\sin(\theta_2) & 0 \end{pmatrix}$$

T_65

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -118 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

T_07

$$\begin{pmatrix} \cos(\theta_5) (\cos(\theta_4) (\sin(\theta_1) \sin(\theta_3) - \cos(\theta_1) \cos(\theta_3) \sin(\theta_2)) + \sin(\theta_4) (\cos(\theta_3) \sin(\theta_1) + \cos(\theta_1) \sin(\theta_2) \sin(\theta_3))) + \sin(\theta_5) (\cos(\theta_4) \\ -\cos(\theta_5) (\cos(\theta_4) (\cos(\theta_1) \sin(\theta_3) + \cos(\theta_3) \sin(\theta_1) \sin(\theta_2)) + \sin(\theta_4) (\cos(\theta_1) \cos(\theta_3) - \sin(\theta_1) \sin(\theta_2) \sin(\theta_3))) - \sin(\theta_5) (\cos(\theta_4) \\ -\cos(\theta_5) (\cos(\theta_2) \sin(\theta_3) \sin(\theta_4) - \cos(\theta_2) \cos(\theta_3) \cos(\theta_4)) - \sin(\theta_5) (\cos(\theta_2) \end{pmatrix}$$

T_76

$$\begin{pmatrix} \cos(\theta_5) & -\sin(\theta_5) & 0 & 0 \\ 0 & 0 & -1 & -290 \\ \sin(\theta_5) & \cos(\theta_5) & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$